

Data-Based Power Coefficient (C_p) Approximation for FOWTs

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Defining the Power Coefficient (C_p)

Theoretical Definition

C_p represents the aerodynamic efficiency of a wind turbine, indicating how much power is extracted from the wind.

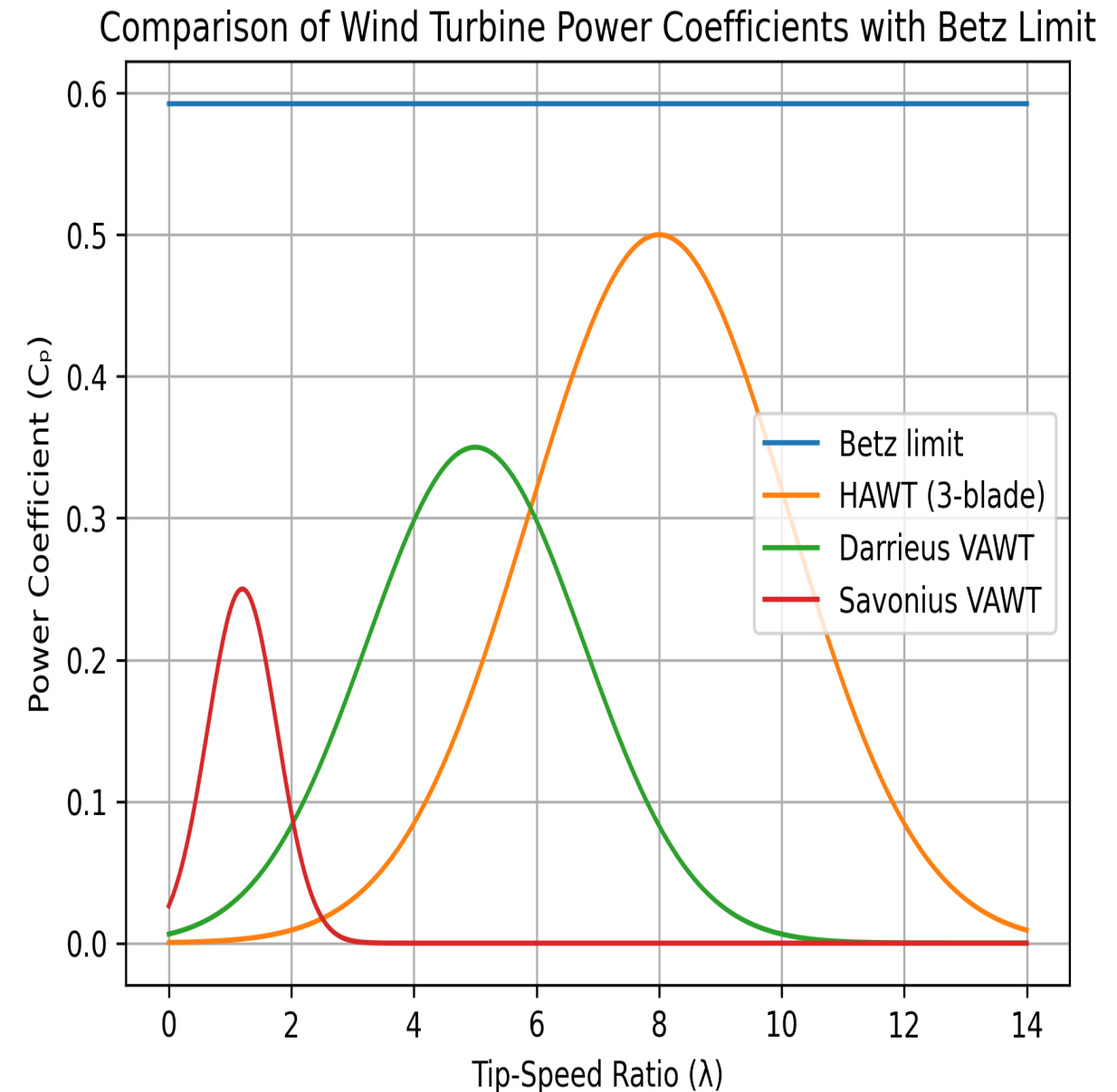
Significance

It's the primary indicator of turbine performance and crucial for Maximum Power Point Tracking (MPPT).

The Betz Limit

Physics limits C_p to a maximum of 0.593 (59.3%).

Power Equation
$$P = \frac{1}{2} \rho A v^3 C_p(\lambda, \beta)$$



Solution : Data-Driven Cp Approximation

Our proposed solution leverages a data-based approach to approximate the power coefficient (C_p), moving beyond fixed analytical models. This methodology learns directly from operational data, offering a more adaptive and accurate C_p estimation.



Machine Learning for Non-Linearity

Captures the complex, non-linear relationship between C_p , tip-speed ratio (λ), and blade pitch angle (β) with high precision.



Implicit Aerodynamic Modelling

Accounts for critical aerodynamic effects such as blade stall, turbulence, and wake interactions through direct data observation.



Enhanced Performance & Reliability

Leads to improved real-time C_p estimation, greater power capture, and more efficient and reliable wind turbine control systems.



Adaptive to Changing Conditions

The model dynamically adjusts to varying wind conditions, ensuring sustained accuracy and performance in diverse environments.

Problem statement

Traditional methods for approximating the power coefficient (C_p) in wind turbines often fall short, relying on static formulas or pre-computed lookup tables. This rigidity prevents them from adapting to the dynamic and complex operating conditions inherent in real-world wind energy generation.



Static & Inflexible Models

Conventional models cannot adjust to varying wind speeds, turbulence, or other environmental shifts, leading to inaccuracies.



Neglect of Complex Aerodynamics

Crucial effects like blade stall, turbulence, and wake interactions are often overlooked, compromising model reliability.



Non-linear Relationship Challenge

C_p is a highly non-linear function of tip-speed ratio (λ) and blade pitch angle (β), which simple analytical models struggle to capture accurately.



Data Generation & Modeling Strategy

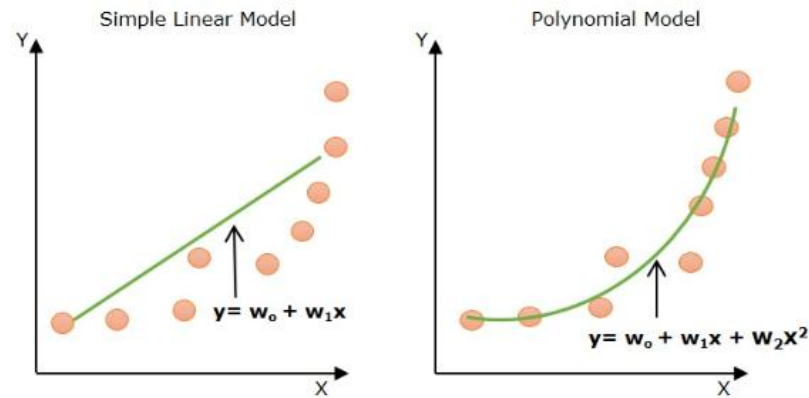
- **The Dataset (Source & Structure)**
 - **Source:** High-fidelity simulations via OpenFAST (NREL).
 - **Features:**
 - **Inputs:** Tip Speed Ratio & Pitch Angle.
 - **Output:** Power Coefficient (Cp).
 - **Scale:** Total Samples: **96,000** (80% Train / 20% Test).
 - **Distribution:** Balanced 50/50 split between **Region 2** and **Region 3**.

6.311485454	0.05729578078	0.4276753453
6.486525513	0.05729578078	0.456336282
6.67176508	0.05729578078	0.4823611693
6.858292464	0.05729578078	0.5188204193
7.0417806	0.05729578078	0.54932415
7.222538565	0.05729578078	0.576490105
7.403141991	0.05729578078	0.6008003397
7.588484584	0.05729578078	0.6233288001
7.773466587	0.05729578078	0.6429588264
7.956954724	0.05729578078	0.6588544492
8.13446739	0.05729578078	0.6704359817

MODELS CONSIDERED

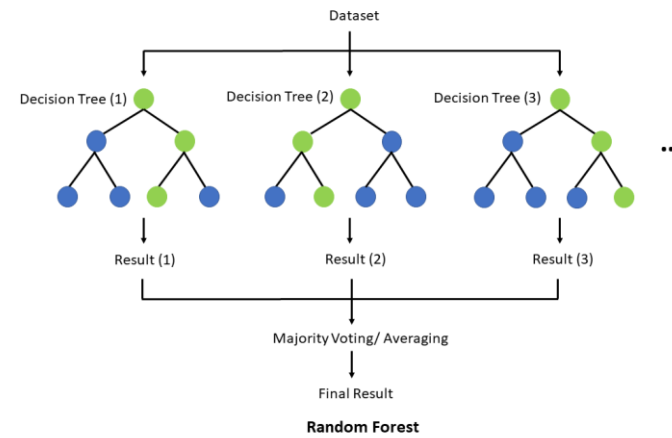
1. Polynomial Regression

- Smooth analytical approximation
- Captures global Cp trends



2. Random Forest Regression

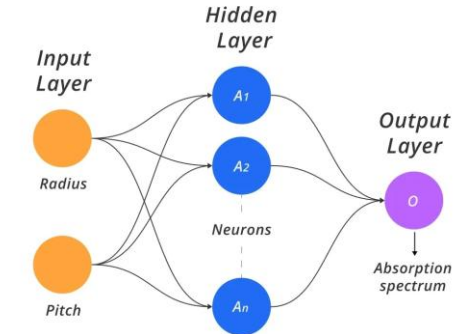
- Tree-based ensemble method
- Robust to noise and local nonlinearities



3. Radial Basis Function (RBF)

- High accuracy for nonlinear mapping
- Effective for smooth, localized data interpolation

Radial Basis Function Network



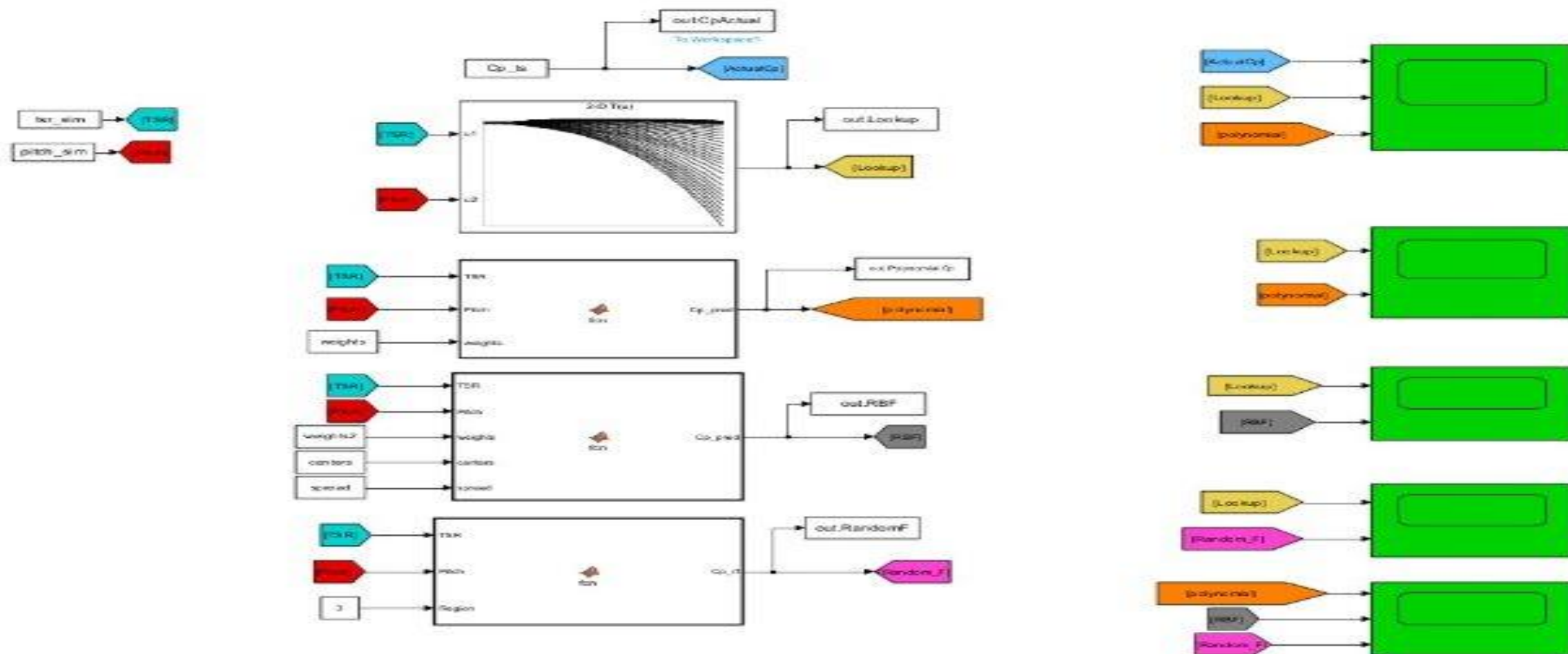
MODEL COMPARISON & KEY INSIGHTS

- **Key Insight**
 - All models show strong generalization with comparable test performance
 - Radial Basis Function achieves higher training accuracy due to improved local adaptability
 - Polynomial regression offers smoother, interpretable trends with limited local flexibility

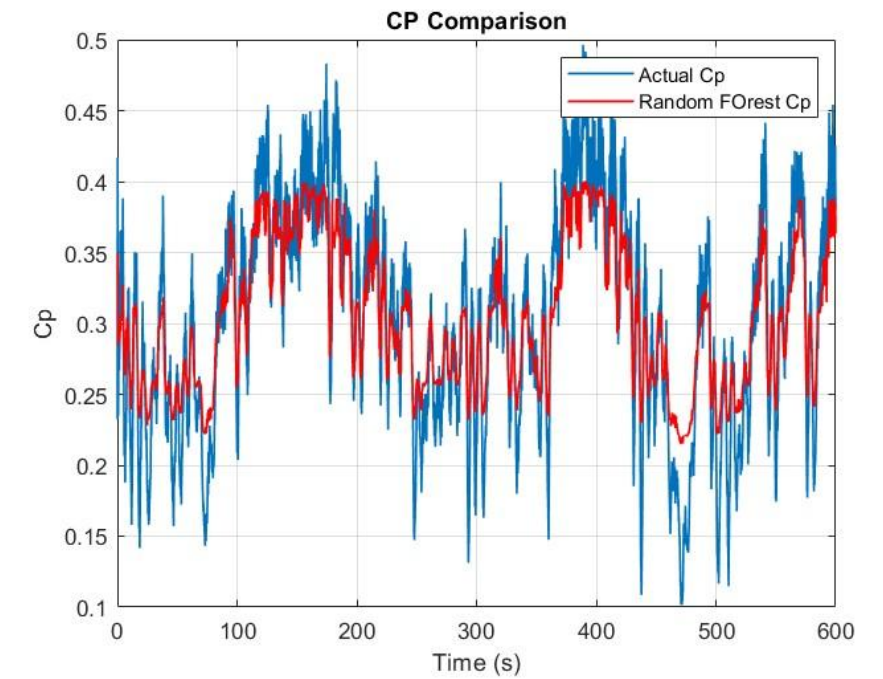
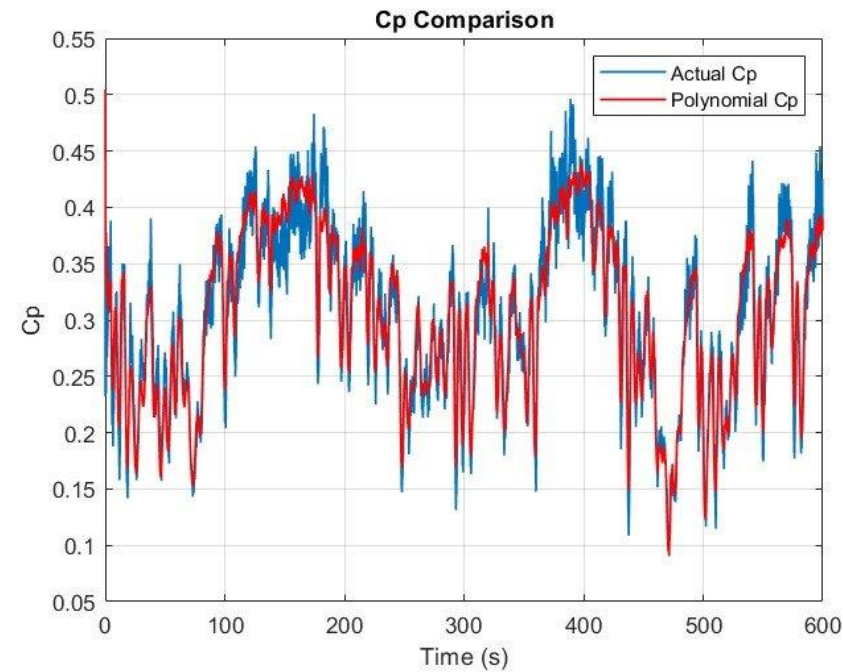
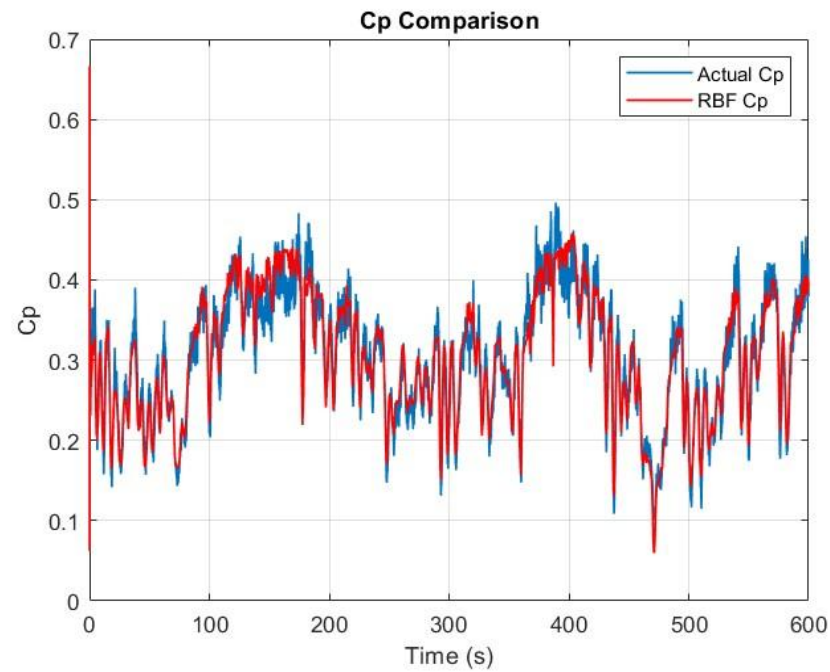
Model	Train R ²	Test R ²	Key Characteristic
Polynomial (Deg-5)	94.99%	95.04%	Smooth global trends, analytical form
Random Forest	97.21%	95.57%	Robust local modeling, nonlinear adaptability
Radial Basis Function (RBF)	97.35	95.89	

Actual vs Predicted Cp (RBF, Polynomial, and Random Forest)

The figure below presents a comparison between the actual power coefficient (C_p) and the values predicted by different trained models, including a fifth-order polynomial regression model, a Random Forest model, and a Radial Basis Function (RBF) network.

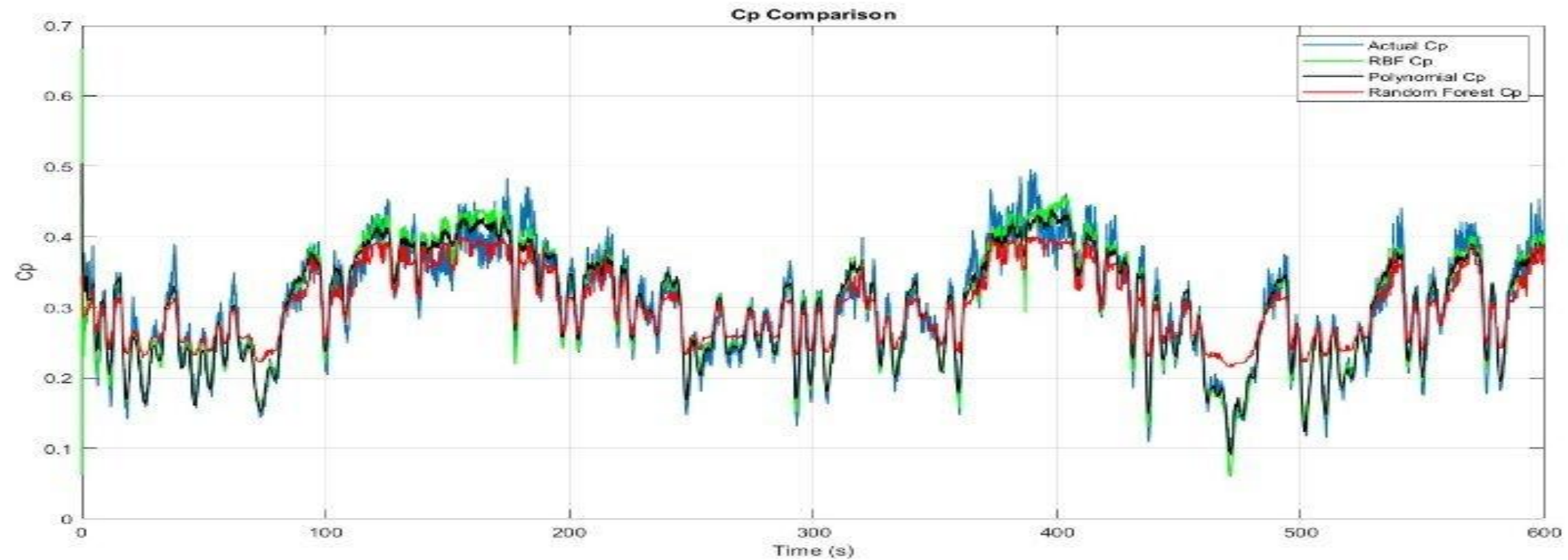


Actual vs Predicted Cp (RBF, Polynomial, and Random Forest)



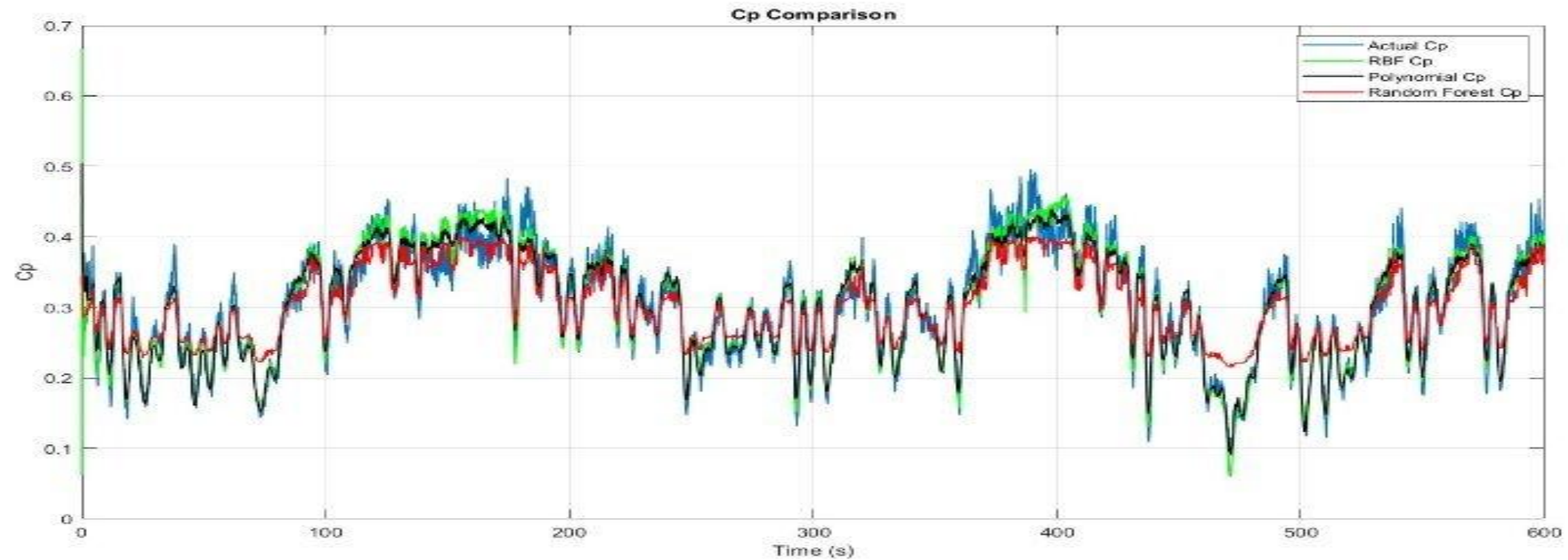
The data-driven models accurately reproduce the temporal variation of the power coefficient, showing good agreement with the actual Cp across the entire operating range and confirming their suitability for dynamic wind turbine modeling.

Actual vs Predicted C_p (RBF, Polynomial, and Random Forest)



The RBF model closely follows the Actual C_p throughout the entire time range, accurately capturing both the magnitude and the rapid fluctuations of the power coefficient, indicating strong dynamic modeling capability.

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Conclusion and Future Work

Conclusion

- The project demonstrated that a non-parametric, data-driven approach can approximate the aerodynamic Power Coefficient (C_p) with very good accuracy.
- By replacing static analytical formulas with a data-based approach, the model effectively captured non-linear aerodynamic behaviors that traditional methods fail to represent.

Future Work

→ **OpenFAST Implementation**

Implement the trained model into a closed-loop OpenFAST controller for final performance testing and real-world application